

Question 1

Suppose in a random experiment, there are two events A and B with the following known probabilities:

- $P(A) = 0.7$
- $P(B) = 0.6$
- $P(A \cap B) = 0.4$

Calculate the following probabilities:

1. $P(A \cup B)$: The probability that A or B occurs.
2. $P(\bar{A} \cap \bar{B})$: The probability that neither A nor B occurs.
3. $P(\bar{A} \cup \bar{B})$: The probability that A and B do not both occur.

Answer:

1. Probability of the Union $P(A \cup B)$

Using the Addition Rule:

$$P(A \cup B) = 0.7 + 0.6 - 0.4 = 0.9$$

Result: 0.9.

2. Probability that neither occurs $P(\bar{A} \cap \bar{B})$

According to De Morgan's Law, "Neither A nor B" is the complement of the union:

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B) = 1 - 0.9 = 0.1$$

Result: 0.1.

3. Probability they do not both occur $P(\bar{A} \cup \bar{B})$

This is the complement of the intersection:

$$P(\bar{A} \cup \bar{B}) = 1 - P(A \cap B) = 1 - 0.4 = 0.6$$

Result: 0.6.

Question 2 (Law of Total Probability)

A genome sequencing center uses three different high-throughput sequencers, which account for 40%, 35%, and 25% of all sequencing reads respectively.

The probabilities that a read contains a sequencing error on each platform are 0.015, 0.010, and 0.020, respectively.

If a sequencing read is chosen at random, what is the probability that it contains a sequencing error?

Answer:

Let A be the event that a sequencing read has an error,

and let B_1, B_2, B_3 represent reads coming from sequencer 1, 2 and 3.

By the Law of Total Probability:

$$P(A) = \sum_{i=1}^3 P(A|B_i)P(B_i) = 0.015 \times 0.4 + 0.010 \times 0.35 + 0.020 \times 0.25 = 0.0145$$

Question 3

A bag contains 1 red ball and 2 white balls. Each time a ball is drawn at random, its color is observed, then the ball is returned to the bag, and **3 additional balls of the same color as the drawn ball** are added to the bag. If we draw a ball from the bag three times in succession, find the probability that the first draw is a red ball, and the second and third draws are both white balls.

Answer

Let:

- A: The event that the first draw is a red ball,
- B: The event that the second draw is a white ball,
- C: The event that the third draw is a white ball.

We need to compute $P(ABC)$. By the general multiplication rule for conditional probability:

$$P(ABC) = P(C|AB) * P(B|A) * P(A)$$

1. Compute $P(A)$:

Initial state: 1 red, 2 white (total = 3 balls).

$$P(A) = 1 / (1+2) = 1/3$$

2. Compute $P(B|A)$:

After drawing a red ball (A), return it and add 3 red balls.

New state: 1+3=4 red, 2 white (total = 4+2=6 balls).

$$P(B|A) = 2 / (4+2) = 2/6$$

3. Compute $P(C|AB)$:

After drawing a white ball (B given A), return it and add 3 white balls.

New state: 4 red, 2+3=5 white (total = 4+5=9 balls).

$$P(C|AB) = 5 / (4+5) = 5/9$$

4. Combine the probabilities:

$$P(ABC) = (5/9) * (2/6) * (1/3) = 10/162 = 5/81$$

Question 4 (sampling with/without replacement).

In a controlled laboratory experiment investigating drug resistance, a researcher works with a finite culture volume containing a total of $N = 10$ bacterial colonies. Within this population, exactly $M = 6$ colonies exhibit resistance to a specific antibiotic. The researcher intends to randomly select a subsample of $n = 2$ colonies for phenotypic testing.

(a) Assume the selection process involves sampling without replacement. Calculate the exact probability that the subsample contains exactly 1 resistant colony.

(b) Suppose the researcher incorrectly treats the selection process as sampling with

replacement, thereby assuming independence between selections. Calculate the resulting probability for observing exactly 1 resistant colony under this incorrect model.

Answer

(a)

All possible event: $n = C_{10}^2 = 45$

Event that subsample contains exactly 1 resistant colony: $n = 6 \cdot 4 = 24$

Therefore, $p = 24/45 = 8/15 \approx 0.53333$

(b)

We define event A: "the sampled colony is resistance colony in one sampling."

$p(A) = 6/10 = 3/5$

Therefore, $p = C_2^1 \cdot p(A) \cdot (1 - p(A)) = 12/25 = 0.48$

Question 5 (Bayes' theorem).

OpenClaw is an AI agent deployed for real-time intrusion detection in a corporate network. It leverages a deep neural network to analyze network traffic and classify each packet as either "malicious" or "benign". Based on historical security logs, the probability that any given packet is malicious is 5%. The neural network has a true positive rate (sensitivity) of 98%—meaning it correctly identifies 98% of malicious packets—and a false positive rate of 3%—meaning it misclassifies 3% of benign packets as malicious. If OpenClaw processes a random packet and outputs a "malicious" classification, what is the probability that the packet is actually malicious?

Answer:

Let M denote the event that a packet is malicious, and T^+ denote a positive (malicious) classification by the neural network. Given:

- $P(M) = 0.05$ (prior probability of malicious traffic)
- $P(T^+ | M) = 0.98$ (sensitivity)
- $P(T^+ | M^c) = 0.03$ (false positive rate)

We want $P(M | T^+)$. Using Bayes' theorem:

$$P(M | T^+) = \frac{P(T^+ | M)P(M)}{P(T^+ | M)P(M) + P(T^+ | M^c)P(M^c)}$$

Substitute the values:

$$P(M | T^+) = \frac{0.98 \times 0.05}{0.98 \times 0.05 + 0.03 \times 0.95} = \frac{0.049}{0.049 + 0.0285} = \frac{0.049}{0.0775} \approx 0.6323$$

Thus, the probability that the packet is truly malicious given a positive classification is approximately **63.23%**.